Thumb to Moon

Calculating the Moon’s diameter

A little window on a large Moon

Let us put the arithmetic aside for a moment. Imagine the Moon as a distant coin. It has a real width, measured in miles, and an apparent width: how much of your view it occupies. A nearby thumb can hide that enormous object because apparent width depends on distance as well as size.

Close one eye and hold your thumb at arm’s length. For a rough picture, think of a thumb about two centimeters—roughly three-quarters of an inch—wide, held about sixty centimeters, or two feet, from your eye. It covers about two degrees. Your own thumb and arm may differ; this is a picture, not a precision instrument. Never use this exercise to look at the Sun.

The Moon spans about half a degree. That is roughly one-quarter of our two-degree thumb width. So a small object covering the Moon at a distance of two feet would be only about a fifth of an inch wide: approximately 0.21 inches, or 5.3 millimeters, at 24 inches.

A correction worth keeping

A TWO-INCH-wide thumb held 24 inches away would cover about 4.77 degrees, not two degrees. Centimeters and inches cannot be exchanged without changing the answer. The rough two-degree example above uses two centimeters, not two inches.

The rule behind the picture

For the SAME apparent width, an object twice as far away must be twice as wide. Imagine sliding a small coin farther from your eye while enlarging it just enough to keep its edges on the same two sightlines. The triangle grows; its shape stays the same. Moving your thumb does not change the Moon. It changes only how much of your view the thumb covers.

Now give the picture a scale

From your arm to the Moon

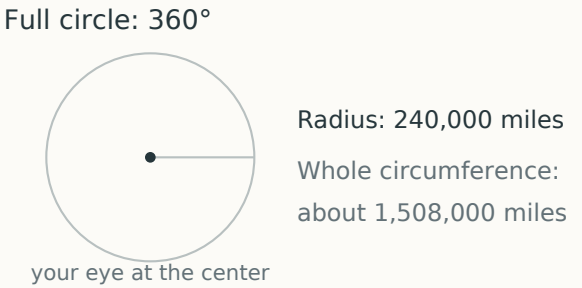
We must already know the distance before we can calculate the Moon's physical width. Take a rounded distance of 240,000 miles. A mile contains 5,280 feet, so that is 1,267,200,000 feet. Divide by our two-foot arm length: the Moon is about 633,600,000 arm lengths away.

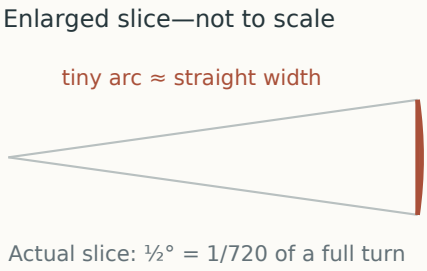
Enlarge the 0.21-inch Moon-covering width by that same factor. The units look cumbersome, but the thought is simple: the same narrow view reaches a much wider object farther away. Converting the enlarged width back into miles gives about 2,100 miles.

An easier route: go around a circle

Imagine an enormous circle centered on your eye, with a radius reaching to the Moon: 240,000 miles. This is an imaginary measuring circle, not a claim about the Moon's orbit. A full turn contains 360 degrees; half a degree is one part in 720.

We must divide the WHOLE circumference by 720—not the radius. The distance around a circle is about 6.28 times its radius. That factor is twice pi; pi is approximately 3.14. Our circle is therefore about 1,508,000 miles around.





Diameter ≈ 240,000 × 6.28 ÷ 720 ≈ 2,100 miles.

The little arc covering half a degree is almost the same length as the straight width across it. At this tiny angle, the difference is negligible for our rough inputs. Dividing the distance alone by 720 would give about 333 miles and omit the essential distance-around-the-circle step.

Does the answer make sense?

Yes. The Moon's known mean diameter is about 2,160 miles, or 3,475 kilometers. Our rounded calculation gives about 2,100 miles, roughly 3,370 kilometers: a good approximation, not an exact measurement. Apparent size changes with lunar distance, and our thumb and arm are rough guides.

We have not discovered the distance with a thumb. We assumed it. Angular width PLUS distance gives physical width. Ancient eclipse geometry offers a different route to size and distance relationships; that is a separate lesson.

Source: NASA Science, Moon Facts • science.nasa.gov/moon/facts/ • Prepared 7 Nov 2026

Geometry is illustrative. Thumb/arm estimates are approximate. Never look at the Sun for this exercise.